

Enterprise Network Design & Security Homelab

Project Report — Multi-Site Simulated Enterprise Network (Cisco Packet Tracer)

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Focus areas	Network Design · Security Hardening · Enterprise Routing
Environment	Cisco Packet Tracer — 3 simulated enterprise sites
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Executive Summary

This project involved designing, building, and securing a simulated enterprise network spanning three interconnected sites in Cisco Packet Tracer. The build was carried out in 18 progressive phases, moving from basic connectivity through to full security hardening — replicating the kind of network an SMB or branch-office deployment would actually run in production. The goal was to demonstrate practical, end-to-end competency across routing, switching, and security configuration, not just isolated exam-style tasks.

Project Scope

Three simulated sites were interconnected to reflect a small multi-branch organization, each with its own local network segments, departmental VLANs, and edge routing back to a central site. The build was structured in 18 phases so each layer of the network — addressing, segmentation, routing, access control, and hardening — was implemented and verified before the next was added.

Core Implementations

VLAN Segmentation — Departmental VLANs configured per site to isolate traffic (e.g. staff, guest, management), reducing broadcast domains and containing lateral movement risk.

OSPF Routing — Dynamic routing configured between sites using OSPF, with route verification and troubleshooting to ensure reliable inter-site connectivity.

NAT / PAT — Network and Port Address Translation configured at site edges to allow internal hosts controlled outbound access while conserving public addressing.

DHCP Services — Centralized and per-VLAN DHCP scopes configured so devices receive addressing automatically and consistently across segments.

Port Security — Switch port security applied to restrict which devices can connect to access ports, mitigating unauthorized device connections.

Access Control Lists (ACLs) — Standard and extended ACLs implemented to enforce traffic policy between VLANs and sites, restricting access to only what's required.

SSH Hardening — Telnet disabled in favor of SSH for all device management, with local authentication and access restrictions configured on routers and switches.

Skills Demonstrated

- Enterprise network topology design across multiple sites
- Layer 2 and Layer 3 troubleshooting and verification
- Security-first configuration mindset (least-privilege access, hardened management)
- Structured, phased project execution and documentation
- Comfort working across routing, switching, and security domains — not siloed to one

Why This Matters for Your Project

This build reflects the same troubleshooting and design thinking needed for real small-business or branch network setups — segmenting traffic properly, keeping remote management locked down, and making sure routing actually converges and stays reliable. If your project involves setting up, securing, or untangling a network, this is the exact skill set applied to get it right the first time.

Full topology diagrams and configuration screenshots available on request.